The strict resultant comparison for the L_α means

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Abstract

We prove the full nondegenerate range of the L_α resultant-mean sign conjecture for both power-mean branches. The proof reduces a mixed derivative to strict concavity of $\sqrt{u}\coth(t\sqrt{u})$ in u. The zero-parameter case is the already known logarithmic-mean equality, which we state explicitly.

Task 1: the original problem

For positive $a \neq b$, the source defines

$$L_\alpha(a,b) = \frac{2\alpha a^\alpha b^\alpha (b-a)}{b^{2\alpha} - a^{2\alpha}}, \qquad L_0(a,b) = \frac{b-a}{\log b - \log a}; \qquad L_\alpha(a,a) = a.$$

These means are symmetric, homogeneous, and satisfy $L_{-\alpha} = L_\alpha$. Let $B_r(a,b) = ((a^r + b^r)/2)^{1/r}$ for $r \neq 0$, and $B_0(a,b) = \sqrt{ab}$. Write

$$\mathcal{R}(B_p, L_\alpha, B_q)(a,b) = B_p\big(L_\alpha(a, B_q(a,b)), L_\alpha(B_q(a,b), b)\big).$$

Conjecture 4.2 of the preprint [1], retained as Conjecture 32 on p. 20 of the complete published article, asks for the negative sign of $L_\alpha - \mathcal{R}$ at the following parameters.

Theorem 1. *Suppose $0 < |\alpha| < 1/2$, and put*

$$r = \sqrt{1 + 16\alpha^2 - 16\alpha^4}, \qquad p = r \text{ or } p = -r, \qquad q = \frac{1-p}{2} - 2\alpha^2.$$

Then, for every distinct positive pair a, b,

$$L_\alpha(a,b) < \mathcal{R}(B_p, L_\alpha, B_q)(a,b).$$

At $\alpha = 0$, the same parameters give equality for all positive inputs. Equality also holds at $|\alpha| = 1/2$ with the limiting parameter pairs, and on the diagonal $a = b$.

The source explicitly records the logarithmic-mean equality at $\alpha = 0$ in the example preceding its conjecture. Thus the printed strict inequality's inclusion of zero is an already known degenerate exception. Our contribution is the proof of the entire nonzero range, not that exception.

A strict-concavity lemma

Lemma 1. *For fixed $t > 0$, the function*

$$\Phi_t(u) = \sqrt{u}\,\coth(t\sqrt{u}), \quad u > 0, \qquad \Phi_t(0) = 1/t$$

is strictly increasing and strictly concave on $[0, \infty)$.

Proof. The classical partial-fraction expansion for coth, [2, Eq. 4.36.3], gives

$$\Phi_t(u) = \frac{1}{t} + 2t \sum_{n=1}^{\infty} \frac{u}{\pi^2 n^2 + t^2 u}.$$

Each summand is increasing and strictly concave. More explicitly,

$$\Phi_t'(u) = 2t \sum_{n=1}^{\infty} \frac{\pi^2 n^2}{(\pi^2 n^2 + t^2 u)^2} > 0, \qquad \Phi_t''(u) = -4t^3 \sum_{n=1}^{\infty} \frac{\pi^2 n^2}{(\pi^2 n^2 + t^2 u)^3} < 0.$$

The differentiated series converge uniformly on compact nonnegative intervals, being bounded by constant multiples of n^{-2} and n^{-4}, respectively. This proves the lemma, including zero by continuity. □

The parameter identity

By symmetry in α, assume $0 < \alpha < 1/2$. Put

$$A = \frac{1}{2}, \quad B = \alpha, \quad \kappa = 1 - 4\alpha^2, \quad C = \frac{|q|}{2}.$$

Here $0 < \kappa < 1$, $1 < r < 2$, and

$$r^2 + \kappa^2 + 2\kappa = 4, \qquad q = (\kappa - p)/2. \tag{1}$$

In particular $pq < 0$. For each $t > 0$, abbreviate $f_c = c\coth(ct) = \Phi_t(c^2)$.

If $p = r$, then $C = (r - \kappa)/4 > 0$, and (1) gives

$$2C^2 = (r+1)B^2 - (r-1)A^2.$$

Thus B^2 is a strict convex combination of C^2 and A^2:

$$B^2 = \frac{2}{r+1} C^2 + \frac{r-1}{r+1} A^2.$$

The points are distinct because $B < A$. Strict concavity gives

$$2f_C < (r+1)f_B - (r-1)f_A. \tag{2}$$

If $p = -r$, then $C = (r + \kappa)/4$, and instead

$$2C^2 = (r+1)A^2 - (r-1)B^2.$$

Applying the same strict concavity with A^2 as the intermediate point gives

$$2f_C < (r+1)f_A - (r-1)f_B. \tag{3}$$

Equivalently, both cases establish the single strict inequality

$$W(t) := (p-1)f_A - (p+1)f_B + 2f_C < 0. \tag{4}$$

From concavity to the resultant mean

Let

$$v(t) = \log \frac{2\alpha \sinh(t/2)}{\sinh(\alpha t)}.$$

Then

$$v' = f_A - f_B > 0$$

by the increasing part of Lemma 1. Since $f_c' = c^2 - f_c^2$, we also have

$$v'' = \frac{\kappa}{4} - (f_A + f_B)v'.$$

Consequently

$$\begin{aligned} T(t) &:= v'' + p(v')^2 + q\coth(qt/2)v' - \frac{\kappa}{4} \\ &= v'\big((p-1)f_A - (p+1)f_B + 2f_C\big) < 0. \end{aligned} \tag{5}$$

Here $q\coth(qt/2) = 2C\coth(Ct) = 2f_C$, for either sign of q.

For $a < b$, define

$$H(a,b) = (b^q - a^q)L_\alpha(a,b)^p, \qquad H(a,a) = 0.$$

Writing $s = p + q$, $t = \log(b/a)$, we obtain

$$H(a,b) = a^s h(t), \qquad h(t) = e^{st/2}\, 2\sinh(qt/2)e^{pv(t)}.$$

Differentiation, using $p + 2q = \kappa$, gives

$$h'' - sh' = phT, \qquad H_{ab}(a,b) = -\frac{a^s}{ab}\, ph(t)T(t) < 0.$$

Indeed h has the sign of q, $pq < 0$, and $T < 0$. It follows, for every $a < c < b$, that

$$H(a,c) + H(c,b) - H(a,b) = \int_a^c \int_c^b H_{ab}(u,v)\, dv\, du < 0. \tag{6}$$

Take $c = B_q(a,b)$, so $c^q - a^q = b^q - c^q = (b^q - a^q)/2$. If $q > 0$, (6) gives

$$L_\alpha(a,b)^p > \frac{L_\alpha(a,c)^p + L_\alpha(c,b)^p}{2}.$$

Now $p < 0$, so taking the p-th root proves $L_\alpha < \mathcal{R}$. If $q < 0$, division by $b^q - a^q$ reverses the displayed inequality, and $p > 0$; again $L_\alpha < \mathcal{R}$. Symmetry covers $a > b$.

The equality cases. At $\alpha = 0$, the parameter pairs are $(p,q) = (1,0)$ and $(-1,1)$. Splitting the logarithmic mean at $c = \sqrt{ab}$ gives

$$\frac{L_0(a,c) + L_0(c,b)}{2} = L_0(a,b).$$

Splitting at $c = (a+b)/2$ gives

$$\frac{L_0(a,c)^{-1} + L_0(c,b)^{-1}}{2} = L_0(a,b)^{-1}.$$

These are the already known equalities cited by the source. At $|\alpha| = 1/2$, $L_\alpha = \sqrt{ab}$ and the parameter pairs are $(2,-1)$ and $(-2,1)$. Direct substitution, using $B_1(a,b)B_{-1}(a,b) = ab$, gives equality. The diagonal equality is immediate from the definitions. This completes Theorem 1. □

Verification and scope. The proof above is entirely analytic. The accompanying verifier checks six exact algebraic identities, and supplements them with 126 strict comparisons and 36 endpoint equalities at 100 decimal digits. Those finite checks audit the formulas and signs; they are not the universal proof. The authoring agent conducted the audit; independent peer review is not claimed.

Proposed follow-up questions

The following four tasks are unproved directions, not additional solves.

Task 2. For each fixed $0 < \alpha < 1/2$, determine the full region of parameter pairs (p, q) for which $L_\alpha < \mathcal{R}(B_p, L_\alpha, B_q)$ holds for every distinct positive input pair.

Task 3. For each of the two parameter branches, determine the sharp constant $K_\alpha > 0$, if it exists, in

$$\frac{\mathcal{R}(B_p, L_\alpha, B_q) - L_\alpha}{L_\alpha} \geq K_\alpha \left(\frac{|\log(b/a)|}{1 + |\log(b/a)|} \right)^6 .$$

Task 4. For $1/2 < |\alpha| < 1$, determine the number and location of sign changes of the same resultant difference as $|\log(b/a)|$ varies.

Task 5. Characterize which homogeneous means with a hyperbolic-product formula for their associated one-variable function admit a resultant comparison deduced from concavity in squared frequencies.

References

[1] L. Mihoković and M. Raïssouli, *Results on comparison and sub/super-stabilizability of some new means*, arXiv:2405.13618v2 (2024), Conjecture 4.2; Applicable Analysis and Discrete Mathematics **20** (2026), 1–23, doi:10.2298/AADM240618001M, Conjecture 32. Complete published text: journal PDF.

[2] NIST Digital Library of Mathematical Functions, Eq. 4.36.3, the classical partial-fraction expansion of the hyperbolic cotangent.